

2 Many of the metabolic processes taking place in cells require ATP.

(a) The electron transport chain (ETC) is involved in the production of ATP.

(i) Where is the site of the electron transport chain?

(1)

- ☐ **A** cytoplasm
- ☐ **B** inner mitochondrial membrane
- ☐ **C** mitochondrial matrix
- ☐ **D** outer mitochondrial membrane

(ii) Enzymes are involved in the production of ATP by the electron transport chain.

Which type of molecule is an enzyme?

(1)

- ☐ **A** carbohydrate
- ☐ **B** lipid
- ☐ **C** phospholipid
- ☐ **D** protein

(iii) Describe the role of reduced NAD in aerobic respiration.

(2)

(b) Carbon monoxide and cyanide affect the production of ATP.

(i) Carbon monoxide binds to the haem group of haemoglobin.

Explain how carbon monoxide will affect the production of ATP.

(3)

(ii) Cyanide stops the transfer of electrons along the electron transport chain.

Explain why cyanide does not affect the production of ATP from anaerobic respiration.

(2)

(Total for Question 2 = 9 marks)
